



Clément Helsens

Senior Data Scientist & ML Engineer
Lake Geneva region · France /
Switzerland

CONTACT

- clement.helsens@epfl.ch
- linkedin.com/in/clementhelsens
- github.com/clementhelsens
- Lake Geneva region, France /
Switzerland

TARGET SECTORS

- Life Sciences — microscopy, biological imaging, bioinformatics
- Medical & Health — medical image analysis, statistical study validation, clinical correlations, data visualisation
- Physics & Engineering — HPC, intensive computing, instrumentation
- Sports & Performance — analytics, predictive models
- Data-intensive R&D — deeptech startups, industry

TECHNICAL STACK

ML & Vision

CoTracker3

ResNet

Deep Learning

Computer Vision

PyTorch

scikit-learn

Statistics

PROFILE

Ph.D. physicist turned senior data scientist & ML engineer with **15+ years** shipping end-to-end data products across High Energy Physics, Life Sciences, and Sports Analytics. Deployed production computer-vision systems (CoTracker3, ResNet), managed **500 TB+** data pipelines, and built interactive analytics platforms used daily by 10+ researchers. Strong background in **Bayesian and frequentist statistical modelling** — directly applicable to clinical study result validation, correlation analysis and medical data visualisation. Author of open-source tooling adopted by multiple international collaborations.

EXPERIENCE

• Independent Contractor

2026 – present

Sports Performance & Nutrition · Self-employed

- **Sports Performance Platform (Django):** full-stack app integrating HRV, continuous glucose monitoring, training logs, nutrition tracking, and a custom race pacing model — pilot with ~10 athletes, targeting ultra-distance events.

Bayesian

Frequentist

Hypothesis testing

ROOT

Data & Infra

MariaDB

Spark

Docker

CI/CD

HTCondor

Slurm

Web & Visualisation

Django

Bokeh

Panel

Programming Languages

Python

C++17

R

SQL

Scientific Instrumentation

Microscopy control

SLIMS

Geant4

Real-time DAQ

EDUCATION

Ph.D. — Particle Physics

Univ. Paris XI & CEA Saclay

2006 — 2009

M.Sc. — Fundamental Physics

Université Paris XI

2003 — 2006

HONORS

2025

Breakthrough Prize —
Fundamental Physics (co-
recipient, ATLAS collaboration)

2013

Higgs boson discovery — co-
author (ATLAS collaboration)

LANGUAGES

FR

Native

EN

Fluent

DE

Intermediate

INTERESTS

Trail running & ultra-distance events,
permaculture, gardening.

- Scientific Data Manager &
R&D Physicist
- 2023 — present
- EPFL · Lausanne, Suisse
- Computer-Vision System (CoTracker3 fine-tuned):** end-to-end pipeline for live biological imaging — sub-second inference, from acquisition through tracking to interactive dashboard. In production for 12+ months. Pre-print submitted.
 - Laser Ablation Control:** full Bokeh dashboard — region selection, laser parameterisation, real-time acquisition control — replacing manual workflows entirely.
 - Drug Screening Pipeline:** automated phenotyping platform (zebrafish) — SLIMS ↔ VAST ↔ Leica integration with 3 ResNet models (orientation, QC, somite counting).
Reduced assay turnaround from days/weeks to minutes.
 - Adaptive Acquisition (Nikon):** two-stage pipeline — cell detection at 10× then optimised imaging path at 40× — sharply increases usable cells per imaging session.
 - Data Platform (500 TB+):** built from scratch a MariaDB single-cell image database (300k+ entries) and a raw-data catalog; currently leading lab-wide tape migration.
 - Researcher Tooling:** Bokeh & Panel analytical dashboards used daily by 10+ scientists across the lab.

• **Research Physicist & Software Coordinator**

2013 – 2022

CERN · Genève, Suisse · Étude FCC & Expérience ATLAS

- **Key4HEP (co-founder):** built an open-source software platform from scratch, now the de-facto standard across multiple major detector projects. Coordinated a 15-researcher international team.
- **Tech Leadership:** ran FCC computing (6 yr) and ATLAS analysis (3 yr) — 50+ researchers across continents, 4 major spec releases, long-term architectural roadmap.
- **Lead Physicist:** led prospect studies for τ -leptonic decays of B^+ and B_c^+ at FCC-ee (first-author JHEP 2021 & EPJC 2024); led the ATLAS top-quark charge-asymmetry measurement at $\sqrt{s} = 8$ TeV (EPJC 2016).
- **Instrument Engineering:** designed and built from scratch an automated X-ray scanning rig (mechanical design, motion control, real-time DAQ software) to characterise a new ATLAS Micromegas muon detector.
- **Public speaking:** ~30 invited talks at international scientific conferences (ICHEP, EPS, SUSY, Higgs Couplings, CHEF, La Thuile).

- **Research Associate** 2009 – 2013

IFAE · Barcelone, Espagne

- **Lead Analyst at IFAE:** pioneered vector-like quark / four-top final-state searches in ATLAS (first-author JHEP 2015) and led precision top-quark pair cross-section measurements at $\sqrt{s} = 7 / 8$ TeV.
- **Statistical Methodology:** introduced and championed profile-likelihood techniques for systematic-uncertainty profiling — adopted broadly across the ATLAS BSM-search programme and codified into reusable open-source tooling.
- **Open-Source Statistical Tooling:** author of two libraries adopted across the international ATLAS community — a C++ frequentist hypothesis-testing library (20+ teams, 15+ peer-reviewed studies) and pyFBU, the first Python multi-channel Bayesian unfolding library (cited in 6+ studies, pioneered PyMC use in the field).
- **Detector Calibration:** designed and deployed end-to-end calibration pipelines across 10 000+ channels of the ATLAS calorimeter.

SELECTED PROJECTS

LiLiTTool

Real-time biological tracking on light-sheet microscopy (EPFL). Fine-tuned

Drug Screen Pipeline

3 ResNet models + SLIMS/VAST/Leica integration for

CoTracker3, sub-second latency, 12+ months in production.

[View project →](#)

automated zebrafish phenotyping. Turnaround reduced from weeks to minutes.

[View project →](#)

Viventis Laser Dashboard

Bokeh dashboard for real-time laser ablation control — ROI selection, laser parameterisation, live visualisation.

[View project →](#)

Sports Performance Platform

Django app: HRV, glucose, nutrition, training & pacing model for ultra-distance athletes. Pilot with ~10 athletes.

[View project →](#)

pyFBU / C++ Stat Tools

Open-source statistical frameworks for CERN — 20+ groups, 20+ publications,

Key4HEP

Open-source software platform for detector physics — co-founder, international standard

pioneered PyMC
use in the field.

[GitHub →](#)

across multiple
major CERN
projects.

[View project →](#)

SELECTED PUBLICATIONS

LIFE SCIENCES · EPFL · 2023-2026

2026 Real-Time Multi-Position and Multi-ROI Tracking with LiLiTTool for Smart Light-Sheet Microscopy in Growing Samples

EPFL preprint · 1er auteur / 1st author

[doi:10.64898/2026.04.03.715281](https://doi.org/10.64898/2026.04.03.715281)

2023 Clock-driven waves of Tbx6 expression prefigure somite boundaries

bioRxiv preprint

[doi:10.1101/2023.11.09.566373](https://doi.org/10.1101/2023.11.09.566373)

2023 Evaluation of genetic demultiplexing of single-cell sequencing data from model species

Life Science Alliance

[doi:10.26508/lsa.202301979](https://doi.org/10.26508/lsa.202301979)

HIGH ENERGY PHYSICS · SELECTED



- 2024** Prospects for B_c^+ and $B^+ \rightarrow \tau^+ \nu_\tau$ at FCC-ee
Eur. Phys. J. C · 1er auteur / 1st author
[doi:10.1140/epjc/s10052-024-12418-0](https://doi.org/10.1140/epjc/s10052-024-12418-0)
- 2021** Prospects for $B^+ \rightarrow \tau^+ \nu_\tau$ at FCC-ee
JHEP 12 (2021) 133 · 1er auteur / 1st author
[doi:10.1007/JHEP12\(2021\)133](https://doi.org/10.1007/JHEP12(2021)133)
- 2019** Heavy resonances at energy-frontier hadron colliders
Eur. Phys. J. C · auteur principal / main author
[doi:10.1140/epjc/s10052-019-7062-3](https://doi.org/10.1140/epjc/s10052-019-7062-3)
- 2016** Measurement of the charge asymmetry in top-quark pair production in the lepton-plus-jets final state at $\sqrt{s} = 8$ TeV with the ATLAS detector
Eur. Phys. J. C · ATLAS Collaboration
[doi:10.1140/epjc/s10052-016-3910-6](https://doi.org/10.1140/epjc/s10052-016-3910-6)
- 2015** Search for production of vector-like quark pairs and of four top quarks in the lepton-plus-jets final state at $\sqrt{s} = 8$ TeV with the ATLAS detector
JHEP 08 (2015) 105 · ATLAS Collaboration · 1er auteur / 1st author
[doi:10.1007/JHEP08\(2015\)105](https://doi.org/10.1007/JHEP08(2015)105)